Strategic Audit Selection and Tax Evasion Detection: An Agent-Based Model Analysis of Risk-Based Targeting with Moore Neighborhood Effects

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ABSTRACT
This paper presents a comprehensive agent-based model (ABM) to investigate the conditional effectiveness of audit selection strategies for detecting tax evasion under conditions of risk aversion and social influence through Moore neighborhood effects. Using NetLogo simulations we find nuanced results that challenge conventional wisdom about audit targeting. With short social learning memory, strategic selection achieves higher fraud detection rates than random and income selection. However, with longer social learning memory, random selection marginally outperforms strategic selection. This conditional effectiveness reveals a critical mechanism where shorter memory prevents adequate risk aversion adjustment, making audit strategy selection paramount; longer memory enables convergence toward high compliance, emphasizing the potential of mixed methods. Our findings align with behavioral economics literature showing that social learning and risk aversion dynamics mediate compliance behavior and support the theoretical prediction that audit strategy effectiveness depends on the interplay between detection strategy, social learning dynamics, and neighborhood effects. These results have important policy implications for tax administration design in contexts where taxpayer behavior is influenced by peer observations and risk perception.

Keywords: Tax evasion, audit selection, agent-based modeling, risk aversion, social learning, Moore neighborhood, detection efficiency, behavioral economics, tax compliance, signaling effects
JEL Classification:

1. INTRODUCTION

1.1 Problem Statement

- Tax evasion remains a critical challenge for tax administrations worldwide
- Limited audit resources require strategic selection of taxpayers to audit
- Three main audit selection strategies: Random, Income-level, Strategic (risk-based)
- Question: Which strategy is most effective? Does it depend on context?

1.2 The Social Learning Memory Gap

- Recent literature emphasizes social learning and neighborhood effects in tax compliance
- **Critical gap:** How does the speed of social learning (memory period) affect audit strategy effectiveness?
- Hypothesis: Social learning memory may moderate which audit strategy works best
- This paper addresses this gap through agent-based modeling

1.3 Research Questions

1. **RQ1:** Does strategic audit selection based on risk scores (PRF) outperform random and income-level selection? (H1)
2. **RQ2:** Does social learning memory (PF) moderate the effectiveness of audit strategies? (H2)
3. **RQ3:** Why does income-level targeting consistently underperform? (H3)
4. **RQ4:** How do audit strategies affect revenue collection beyond fraud detection rates? (H4)

1.4 Contribution

- First ABM study examining how social learning memory modulates audit strategy effectiveness
- Demonstrates that strategy choice depends on social learning capacity in the population
- Provides policy guidance for tax administrations with different compliance environments

2. LITERATURE REVIEW

2.1 The Insufficiency of Deterrence Theory and the Necessity of Heterogeneity.

Tax evasion research traces its formal origins to the application of expected utility theory under uncertainty. The foundational deterrence model (Allingham and Sandmo, 1972) conceptualized evasion as a rational gamble: taxpayers weigh the certain cost of paying taxes against the risky prospect of evading, where detection triggers penalties. Under this framework, three mechanisms govern compliance—audit probability, penalty severity, and individual risk aversion. Increasing enforcement intensity or penalties should reduce evasion; taxpayers with higher risk aversion should declare more income.

Yet persistent empirical anomalies revealed the model's fundamental inadequacy. Observed compliance rates systematically exceeded theoretical predictions under realistic enforcement parameters. Where third-party information reporting exists, compliance approaches near-perfection; where such reporting is absent, substantial underreporting occurs—yet even this underreporting remains far below what pure deterrence theory predicts given audit probabilities

typically below ten percent (Slemrod, 2007). These patterns indicated that taxpayers were not homogeneous rational actors responding identically to enforcement parameters, but rather heterogeneous individuals whose compliance reflected intrinsic motivations ("tax morale"), social norms, and contextual factors that deterrence theory could not accommodate.

The recognition that heterogeneity matters fundamentally—that taxpayers differ not just in risk preferences but in decision-making processes themselves—created a methodological difficulty. Traditional analytical approaches built on representative agents could not capture populations where some individuals always comply due to ethical commitments, others adjust strategically based on enforcement experiences, and still others evade reflexively regardless of consequences. Moreover, if compliance decisions depend on beliefs formed through social interactions, and if these beliefs evolve endogenously as taxpayers observe neighbors' experiences with tax authorities, then the static equilibrium analysis characteristic of early theoretical models becomes limited.

2.2 Agent-Based Modeling as Methodological Solution: Archetypes, Learning, and Emergence.

Agent-based modeling emerged as the methodological response to deterrence theory's empirical failures. Rather than assuming representative agents and solving for equilibria, ABM constructs populations of heterogeneous individuals, specifies their decision rules, embeds them in interaction structures, and observes what aggregate patterns emerge from micro-level dynamics (Hokamp et al., 2017). This bottom-up approach offered three critical advantages for tax compliance research. First, it permitted multiple decision-making processes to coexist within the same population. Second, it allowed beliefs and attitudes to update endogenously through reinforcement learning and social observation. Third, it enabled controlled computational experiments comparing policy interventions—particularly audit strategies—that would be infeasible to test in natural settings.

The translation of heterogeneity into operational models required developing taxonomies of decision-making types. Behavioral archetype frameworks emerged as the solution, categorizing taxpayers not by continuously varying parameters but by qualitatively distinct decision processes. The most influential typology distinguished four archetypes: agents who always comply due to internalized norms (ethical or honest types); agents who adjust behavior through reinforcement learning, increasing compliance after being audited and decreasing it after avoiding detection (strategic types); agents who evade unless enforcement exceeds a calculated threshold (defiant types); and agents whose declarations reflect unintentional errors rather than deliberate optimization (random types) (Bloomquist, 2011; Hokamp and Pickhardt, 2010).

Embedding these archetypes within evolutionary frameworks revealed dynamics invisible to static models. When agent fitness depends on post-tax, post-penalty income, and when less-fit agents exit the population at higher rates, evolutionary pressures favor evasion: successful evaders survive longer, gradually displacing honest taxpayers from the population (Bloomquist, 2011). This dynamic explains why compliance can erode over time even without changes in formal enforcement—the composition of the taxpayer population itself evolves. Critically, the rate of this

evolutionary process depends on how quickly information about enforcement actions propagates through the population, connecting archetype dynamics to social learning mechanisms.

Extensions incorporating temporal depth demonstrated that enforcement looking backward matters as much as enforcement looking forward. When audit regimes cover multiple previous tax years rather than just current declarations—what researchers term "lapse of time" effects—evasion drops substantially because agents must consider cumulative risk across periods (Hokamp and Pickhardt, 2010). This finding has profound implications: it suggests that creating long memory of enforcement experiences, whether through institutional design (multi-year audits) or through social transmission (agents learning about neighbors' past audit experiences), may deter more effectively than increasing current-period audit intensity. Yet existing research examined institutional memory (formal audit scope) without connecting it to social memory (the temporal window of neighborhood observations that influence behavior).

2.3 Social Networks, Neighborhood Dynamics, and the Strategic Problem of Audit Selection

The recognition that taxpayers are embedded in social networks where information about enforcement experiences transmits between agents fundamentally altered how researchers conceptualized compliance dynamics. Rather than responding solely to their own audit experiences, taxpayers observe neighbors' experiences and update their subjective probability assessments accordingly.

Social network models reveal a fundamental tension in tax compliance: while a potential mechanism exists whereby observing neighbors' successful evasion without detection could reduce perceived audit risk and encourage imitation (visible in computational results when enforcement decreases), maintaining compliance under weak enforcement paradoxically requires the opposite—that taxpayers largely refrain from acting on information about the successful payoffs of their neighbors' evasion (Korobow et al., 2007). This finding suggests that social norms actively discouraging attention to the 'gains' of evaders may be as critical to voluntary compliance as formal enforcement institutions themselves

Empirical evidence suggests these social mechanisms can either reinforce or undermine enforcement efforts depending on context. The mechanism operates through contagion: pockets of noncompliance spread spatially when neighbors observe successful evasion. Moreover, the effect of social influence proves non-monotonic.

Under conditions of weak audit probability and low penalties, social contagion—operationalized as agents' convergence to the median evasion behavior of their social network—paradoxically enhances aggregate compliance by dampening the rapid escalation of evasion. However, when enforcement strengthens through increased audits or penalties, this same mechanism becomes a compliance liability: agents anchored to peer behaviors fail to optimize their responses to stronger deterrence signals (Noguera et al., 2014). This suggests that tax authorities must account for how social network effects mediate the effectiveness of enforcement policy changes.

Network topology—how agents are spatially arranged and who observes whom—critically determines the speed and pattern of information propagation. The standard Moore neighborhood specification (eight neighbors surrounding each agent on a grid) represents a methodological compromise between computational tractability and realistic social network representation. Yet empirical calibrations reveal that the appropriateness of neighborhood influence varies across institutional contexts: while some populations exhibit strong neighborhood effects, others show no evidence that taxpayers condition behavior on neighbors' compliance (Arslan and Ican, 2013; Bloomquist, 2011), suggesting that cultural and institutional factors moderate the strength of social learning mechanisms.

When taxpayers update their audit beliefs through repeated experience and social learning, the predictability of audit distribution across population segments becomes critical. Heterogeneous audit strategies that concentrate enforcement differently in different periods create dynamic instability: the compliance losses from periods of weak enforcement pressure exceed the gains from periods of intense pressure (Hashimzade et al., 2014). Strategies maintaining uniform audit intensity across occupational groups—rather than concentrating audits narrowly or varying them randomly—produce higher and more stable tax revenues by avoiding this asymmetric compliance response.

2.4 The Critical Gap: Social Learning Memory as Unexamined Moderator

The literature has established four foundation stones. First, taxpayers are heterogeneous in decision-making processes, not just parameters. Second, these heterogeneous agents are embedded in social networks where information about enforcement transmits between neighbors. Third, the speed and pattern of information propagation depends on network topology. Fourth, different audit selection strategies generate varying detection rates and revenue outcomes, with signaling effects undermining strategies that create predictable audit patterns.

Yet these insights remain disconnected from a fundamental question: How does the temporal extent of social learning—the duration for which neighbors' audit experiences remain behaviorally relevant—moderate which audit selection strategy most effectively deters evasion? This question matters because social learning is not instantaneous but accumulates over time. In populations with short social memory, agents respond primarily to very recent neighborhood audit events, potentially making which specific taxpayers are audited (strategic selection) critical for maintaining deterrence across the population. Conversely, in populations with long social memory, historical audit information accumulates in agents' subjective probability assessments over many periods, potentially allowing compliance norms to stabilize through distributed social learning regardless of whether current audits target high-risk evaders or random taxpayers.

Existing research has examined social learning mechanisms (Korobow et al., 2007; Noguera et al., 2014; Alvarado, 2019) and compared audit strategies (Hashimzade and Myles, 2014; LLacer et al., 2012) but never tested whether social learning memory affect audit strategy effectiveness. Research demonstrating powerful lapse of time effects examines institutional audit scope rather than social memory duration (Hokamp and Pickhardt, 2010). Investigations showing social

contagion can complement or substitute for formal enforcement do not examine whether these interactions depend on how quickly social information updates (Noguera et al., 2014). The consequence is a critical knowledge gap: we cannot determine whether optimal audit policy should vary across jurisdictions based on taxpayer populations' social learning characteristics.

This gap has direct policy implications as tax administrations increasingly adopt risk-based targeting enabled by data analytics and machine learning. If strategic targeting's advantage depends on taxpayers having short social memory—such that only directly audited agents meaningfully update behavior—then investments in sophisticated risk scoring may be unnecessary in populations with robust social learning where random audits achieve similar deterrence by distributing enforcement signals that propagate broadly through networks. Conversely, if strategic targeting maintains advantages even with long social memory, this justifies continued development of predictive algorithms. Without evidence on how social learning memory moderates' strategy effectiveness, policy lacks empirical grounding.

Moreover, the mechanism explaining why income-level targeting underperforms remains underspecified. The hypothesis is that concentrating audits on high-income taxpayers' signals safety to low-income groups, but if this operates through social learning—low-income agents observing their low-income neighbors rarely face audits—then how quickly this perverse signal propagates should depend on social memory duration. Similarly, the disconnect between fraud detection rates and revenue collection documented in prior research (strategic selection can generate substantially higher revenue despite only marginally higher detection rates) suggests targeting matters more for revenue than detection, but whether this holds across different social learning environments remains untested.

A bibliometric synthesis of 978 documents on tax evasion over seven decades reveals a critical integration gap: research examining interactive effects between social learning dynamics and enforcement strategies remains compartmentalized across distinct thematic clusters encompassing behavioral economics, social network effects, and audit selection design (Ahmi et al., 2023). Although scholarship within each cluster has produced theoretically and empirically sophisticated contributions, the literature lacks an integrated framework synthesizing these dimensions to address the central policy question.

This fragmentation engenders direct policy implications. Without empirical evidence regarding how social learning propagation speed and durability moderate the comparative effectiveness of strategic targeting versus probabilistic audit allocation, resource allocation decisions rest on uncertain foundations. Should taxpayer populations demonstrate limited temporal persistence of audit-related information in social networks—such that behavioral responses concentrate among directly audited agents—risk-based targeting strategies warrant their computational and administrative overhead. Conversely, should populations exhibit extended social learning windows—wherein audit information propagates extensively through networks irrespective of which taxpayers face selection—probabilistic audit strategies may achieve equivalent compliance outcomes while substantially reducing resource requirements for predictive analytics infrastructure

Addressing this gap requires testing four interrelated propositions. First, under short social learning memory, strategic risk-based targeting should substantially outperform random and income-level selection because only precision targeting of high-risk evaders maintains deterrence when enforcement signals decay quickly. Second, under long social learning memory, strategic targeting's advantage should diminish as historical audit information accumulates broadly through networks, allowing random selection to achieve similar deterrence at lower administrative cost. Third, income-level targeting should consistently underperform regardless of memory duration because its predictable pattern signals safety to low-income groups—but the magnitude of underperformance should depend on how quickly this signal propagates. Fourth, revenue and detection represent distinct objectives responding differently to strategies: strategic targeting may maximize revenue by detecting fraud among high-income taxpayers who pay larger penalties, but this advantage may depend on social learning preventing compliance improvement that would reduce the pool of high-income evaders to catch.

Testing these propositions through computational experiments varying social learning memory systematically while comparing audit strategies will provide the first evidence on whether optimal enforcement policy should be conditional on populations' social learning characteristics—a fundamental question for tax administration in an era of increasing analytical sophistication and persistent resource constraints.

3. MODEL DESCRIPTION

Entiendo Ariel debe hacer esto: tomar en cuenta los nombres de las salidas que usé en las secciones 4 y 5 y usar esos en vez de los que tenemos en el documento preliminar.

4. EXPERIMENTAL DESIGN AND METHODS

Given that the model has many parameters that we can set, we focused on only the ones that will allow us analyzing the stated research questions and hypotheses. Thus, we used a full factorial 3^1 2^1 design (one factor at three levels and one factor at two levels), where the first factor is the method used to select the taxpayers who will be audited and the second factor is the number of passed periods of neighbors who might have committed a tax fraud (i.e., who might have underreported their respective income) of each taxpayer. Each taxpayer will consider this information to adjust its risk aversion level, as previously stated in the description of the model. More formally, the experimental design is the following:

Factor A: tax-payer-audit-selection: a_1 = strategic a_2 = random a_3 = income-level

Factor B: period-frauds: $b_1 = 1$ $b_2 = 5$

Each of the 6 treatment combinations was run 5 times, so that we can get an approximate 0.95 statistical power at $\alpha = 0.05$.

The measured responses (outputs) of the model were collected at the time the model was clearly in steady state. This was previously visually assessed by inspecting the outputs of several runs of the model under varying experimental conditions. Fig. XN shows a representative graph of outputs that clearly shows that the model enters steady state at about time-step = 10.000. Thus, to be on the safe side, we decided to measure the following responses between times-steps = 25.000 and 30.000:

R1: Mean risk aversion level of taxpayers (MRA) calculated between times-steps = 25.000 and 30.000.

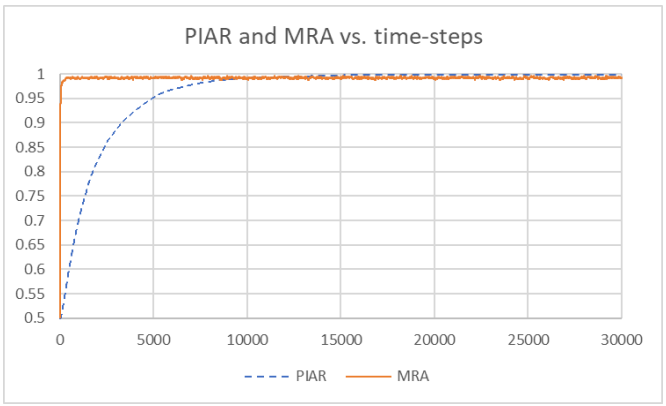
R2: Mean percent of income actually reported by taxpayers (PIAR) between times-steps = 25.000 and 30.000.

R3: Percent of detected frauds (PDF): calculated as cumulative number of frauds detected divided by cumulative total number of frauds committed by taxpayers between times-steps = 25.000 and 30.000.

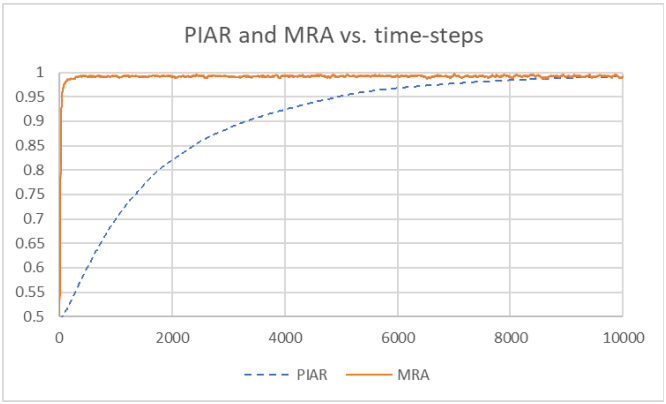
R4: Cumulative difference between real income and reported income of taxpayers (CDI), who were audited and if they had committed fraud (underreport their income), they would have been detected. This difference was calculated and accumulated between times-steps = 25.000 and 30.000.

The other parameters of the model (inputs) were held constant at the following values (see description of the model for an explanation of each parameters' label):

nr-tax-payers = 500
learning-rate = 0.05
nr-auditors = 20
actual-audit-rate = 0.20
tax-rate = 0.27
penalty-rate = 1.5
initial risk-aversion level: 0.5
threshold-influence-income: 1.65



(a)



(b)

Figure XN. Mean risk aversion level (MRA) and Mean percent of income actually reported by taxpayers (PIAR). Panel (a): for 30.000 time-steps, panel (b): for 10.000 time-steps.

5. RESULTS

After the five runs for each of the 6 treatment combinations were done, we calculated the mean of each of the responses. In the following subsections we will present and analyze each of the responses. We must note that in this section, we present and analyze the data focusing on a statistical analysis and explaining the raw findings. A pragmatic/substantive analysis and discussion of those results will be done in the Discussion section.

5.1 Mean risk aversion level of taxpayers (MRA)

Fig. X1 shows the MRA, where we can see that there exists a statistically significant difference in MRA between PF (period-frauds) = 5 and 1 for all tax-payer-audit-selection methods (all p-values of corresponding t-stats < .0001). The MRA for PF = 5 is bigger than for PF = 1. This happens because when taxpayers have a much longer memory to see whether their neighbors have committed a tax

fraud and have been detected (PF = 5), they will increase their risk aversion level more than if they have a shorter memory (PF = 1), i.e., having a longer memory increases the probability that a taxpayer might perceive that one of its neighbors has committed a fraud and has been detected, which makes the taxpayers more cautious.

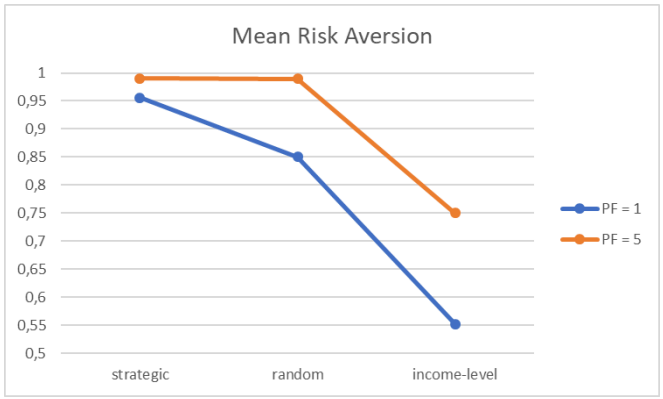


Figure X1. Mean risk aversion level of taxpayers

Now, analyzing the MRA for PF = 5 for each of the tax-payer-audit-selection methods, we can see that the difference in MRA between the strategic and random selection of taxpayers (0.9903 – 0.9891 = 0.0012) is rather small and not statistically significant (p-values of corresponding t-stat = 0.308). This means that if taxpayers have a rather long memory (PF = 5), their risk aversion will be quite the same for these two tax-payer-audit-selection methods. This happens because with a long memory, taxpayers adjust their risk aversion level more effectively according to the general behavior of the rest and thus, having a more or less sophisticated tax-payer-audit-selection method is not so important. On the other hand, if we focus on auditing the higher income taxpayers (income-level tax-payer-audit-selection method), the difference in MRA is statistically significant compared with strategic and random selection of taxpayers (0.9903 – 0.7498 = 0.2405 and 0.9891 – 0.7498 = 0.2393, p-values of t-stats < .0001). In this case, given that we focus on auditing only the higher income taxpayers, the rest of them perceive that the probability of being audited is quite small and thus, they will not increase too much their risk aversion. The same pattern can be seen for PF = 1 (shorter memory), but with some variations. For PF = 1, the difference in MRA between strategic and random tax-payer-audit-selection method is statistically significant (0.9556 – 0.8504 = 0.1052, with p-value of t-stat < .0001). Given that taxpayers have a shorter memory, they will not properly adjust their risk aversion by this process and hence, a more fine-tuned tax-payer-audit-selection method (strategic) will have a higher impact on MRA than a less sophisticated one (random).

To formalize those findings, Table X1 shows an ANOVA of MRA, where we can see that its results are consistent with the previous analyses. The main effects of tax-payer-audit-selection and period-frauds are statistically significant as well as their interaction.

Table X1. ANOVA for Mean risk aversion level of taxpayers (MRA) corresponding to the full factorial experiment with 5 replications.

	Type III SS	df	MS	F	p-value
Corrected model	0.744	5	0.149	3385.224	< .0001
Intercept	21.568	1	21.568	490359.7	< .0001
tax-payer-audit-selection	0.596	2	0.298	6772.2	< .0001
period-frauds (PF)	0.115	1	0.115	2608.1	< .0001
tax-payer-audit-selection x PF	0.034	2	0.017	386.8	< .0001
Error	0.001	24	4.398×10^{-5}		
Total	22.314	30			

5.2 Mean percent of income actually reported by taxpayers (PIAR)

As expected, the behavior of PIAR is rather similar to the one of MRA, as Fig. X2 shows. Remember that taxpayers adjust PIAR according to their risk aversion level.

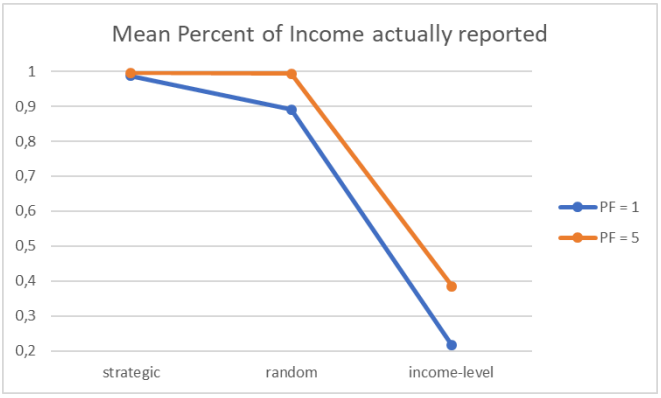


Figure X2. Mean percent of income actually reported by taxpayers

PIAR is always bigger for PF = 5 than for PF = 1. Although that difference is small for strategic tax-payer-audit-selection method (0.9962 – 0.9865 = 0.0097), the p-value for the t-stat is .029. PIAR is always bigger for PF = 5 than for PF = 1 for the same reason previously explained for MRA. Given that a longer memory allows taxpayers to better adjust (increase) their risk aversion; in turn, this permits a better adjustment of PIAR, i.e., risk aversion increases and accordingly so does PIAR.

Now, analyzing PIAR for PF = 5 for each of the tax-payer-audit-selection methods, we can see that for strategic, the difference in PIAR compared with random (0.9962 – 0.9928 = 0.0034) is not statistically significant (p-value of t-stat = 0.063), whereas PIAR is statistically significant smaller for income-level (all p-values of t-stats < .0001). This happens because with a long memory, taxpayers adjust their risk aversion level more effectively according to the general behavior of the rest and thus, having a more or less sophisticated tax-payer-audit-selection method is not so important. On the other hand, if we focus on auditing the higher income taxpayers (income-level

tax-payer-audit-selection method), the difference in PIAR is statistically significant compared with strategic and random selection of taxpayers (0.9962 – 0.3848 = 0.6114 and 0.9928 – 0.3848 = 0.608, p-values of t-stats < .0001). In this case, given that we focus on auditing only the higher income taxpayers, the rest of them perceive that the probability of being audited is quite small and thus, they will not increase too much their risk aversion and accordingly they will consistently underreport their income. The same pattern can be seen for PF = 1 (shorter memory), but with some variations. For PF = 1, the difference in PIAR between strategic and random tax-payer-audit-selection method is statistically significant (0.9865 – 0.8908 = 0.0957, with p-value of t-stat < .0001). Given that taxpayers have a shorter memory, they will not properly adjust their risk aversion by this process and hence, a more fine-tuned tax-payer-audit-selection method (strategic) will have a higher impact on risk aversion than a less sophisticated one (random). In turn, this difference in risk aversion adjustment will have an impact on PIAR, where a more sophisticated tax-payer-audit-selection method (strategic) will allow more fine-tuned audits than a less advanced method (random). Finally, income-level tax-payer-audit-selection method has the worst performance for the same reasons already explained for MRA.

Table X2 shows an ANOVA for PIAR, where we can see that its results are consistent with the previous analyses. The main effects of tax-payer-audit-selection and period-frauds are statistically significant as well as their interaction.

Table X2. ANOVA for Mean percent of income actually reported by taxpayers (PIAR) corresponding to the full factorial experiment with 5 replications.

	Type III SS	df	MS	F	p-value
Corrected model	3.070	5	0.614	2389.1	< .0001
Intercept	16.626	1	16.626	64692.6	< .0001
tax-payer-audit-selection	2.972	2	1.486	5782.1	< .0001
period-frauds (PF)	0.066	1	0.066	256.8	< .0001
tax-payer-audit-selection x PF	0.032	2	0.016	62.3	< .0001
Error	0.006	24	.000257		
Total	19.702	30			

5.3 Percent of detected frauds (PDF)

Before analyzing this response, we must note that PDF was calculated as the total number of frauds detected divided by the total number of frauds committed by all taxpayers between times-steps = 25.000 and 30.000. Given that we set the percent of taxpayers to be audited (actual-audit-rate) equal to 20%, a maximum of 20% of all frauds could be detected. That is why in Fig. X3, PDF reaches a maximum of 18.5%.

As Fig. X3 shows, PDF is bigger for PF = 1 than for PF = 5 only for strategic tax-payer-audit-selection method (0.1848 – 0.1688 = 0.016, with p-value of t-stat < .001). For random and income-level tax-payer-audit-selection methods that difference is not statistically significant (0.1734 – 0.1732 = 0.0002, p-value of t-stat = 0.656; 0.1108 – 0.1080 = 0.0028, p-value of t-stat = 0.444, respectively).

This happens because the most advanced tax-payer-audit-selection method (strategic) provides a better, more fine-tuned selection of taxpayers to be audited, resulting in higher PDF. That difference is notably bigger for PF = 1 than for PF = 5, because for PF = 1 taxpayers adjust risk aversion and thus also PIAR less efficiently than when PF = 5, which makes fraud detection easier.

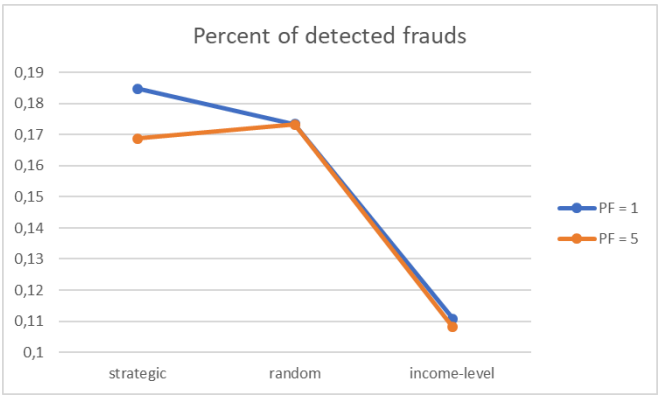


Figure X3. Percent of detected frauds

Analyzing PDF for PF = 5, we can see that for random tax-payer-audit-selection method it is bigger than for strategic (0.1732 – 0.1688, p-value of t-stat = .014). A plausible explanation for this result is that a more advanced tax-payer-audit-selection method (strategic) over focus the audits, and given that a longer taxpayer memory adjusts MRA and PIAR well, this over focusing misses including taxpayers that actually commit frauds. Instead, random tax-payer-audit-selection method is more neutral and thus, it does not suffer from over focusing audits. For strategic and random tax-payer-audit-selection method, PDF is always bigger than for income-level (0.1688 – 0.1080 = 0.0608; 0.1732 – 0.1080 = 0.0652; both p-values of t-stats < .0001) for the same reasons already explained for PIAR: consistently focusing only on the higher income taxpayers, makes the rest of taxpayers care less about underreporting income. This effect combined with a narrowly focused audit selection method (income-level), decreases the probability of detecting frauds. The same pattern is observed for PF = 1, but in this case, strategic is always the best tax-payer-audit-selection method, even compared with random (0.1848 – 0.1734 = 0.0114, p-value of t-stat = .001). A more advanced audit selection method combined with a short taxpayer memory, allows a better targeting of taxpayers that might commit frauds because they care less about underreporting their income.

Table X3 shows an ANOVA for PDF, where we can see that its results are consistent with the previous analyses. The main effects of tax-payer-audit-selection and period-frauds are statistically significant as well as their interaction.

Table X3. ANOVA for Percent of detected frauds (PDF) corresponding to the full factorial experiment with 5 replications.

	Type III SS	df	MS	F	p-value
Corrected model	0.029	5	0.006	380.06	< .0001

Intercept	0.704	1	0.704	45423.4	< .0001
tax-payer-audit-selection	0.029	2	0.014	928.96	< .0001
period-frauds (PF)	< 0.000001	1	< 0.000001	19.10	< .0001
tax-payer-audit-selection x PF	< 0.000001	2	< 0.000001	11.63	< .0001
Error	< 0.000001	24	1.549×10^{-5}		
Total	0.734	30			

5.4 Cumulative difference between real and reported income of taxpayers (CDI)

One concern that the analyses of PDF brings about is that a better audit targeting of taxpayers could increase PDF, but may not be the most effective means of increasing tax revenue. From an egalitarian perspective, increasing PDF, and thus detecting and penalizing fraudulent taxpayers is good. However, from an economic/pragmatic point of view, what matters is how much more tax amount is collected. Hence, perhaps using an income-level tax-payer-audit-selection method could collect more tax amount than the other two methods, although PDF will be smaller, as previously seen.

To assess this question, we measured the difference between the real income of all taxpayers and the actual income reported, when a fraud was detected. That difference was calculated at each time step of a simulation run for all taxpayers and accumulated between times-steps = 25.000 and 30.000, i.e., well after the model entered steady state. Fig. X4 shows the mean of that difference (CDI) across the 5 replications.

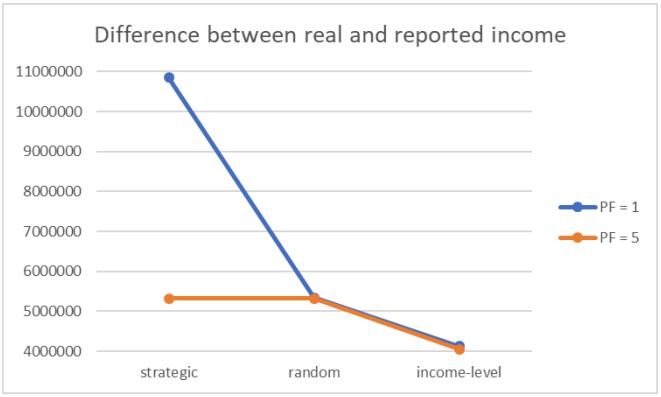


Figure X4. Cumulative difference between real and reported income of taxpayers (CDI)

As expected, Fig. X4 resembles the form of Fig. X3, given that PDF moderates mediates the difference between real and reported income of taxpayers (CDI), which only applies when a fraud is detected. Note that the difference in PDF between PF = 1 and PF = 5 is also reflected in CDI, where CDI is much bigger for PF = 1 than for PF = 5 when we use strategic tax-payer-audit-selection method (10,851,931 – 5,322,423 = 5,529,508, with p-value of t-stat < .001). On the other hand, for random and income-level strategic tax-payer-audit-selection methods, that difference in CDI is not

statistically significant (5,342,515 – 5,324,569 = 17,946 with p-value of t-stat = .656; 4,131,763 – 4,053,318 = 78,445 with p-value of t-stat = .444). This means that for strategic tax-payer-audit-selection method, its bigger PDF compared with random and income-level offsets the smaller tax revenue that strategic might generate compared with random, and especially with income-level tax-payer-audit-selection method.

For PF = 5, the difference in CDI is not statistically significant between strategic and random tax-payer-audit-selection method (5,322,423 – 5,324,569 = -2,145, p-value of t-stat = 0.968), which suggests that PF = 5 allows a good adjustment of MRA and PIAR, making these two methods similar in their PDF and thus, also alike in CDI. On the other hand, the difference in CDI between those two methods and income-level tax-payer-audit-selection is statistically significant (5,322,423 – 4,053,318 = 1,269,105; 5,324,569 - 4,053,318 = 1,271,251; both with p-values of t-stats < .00001). Thus, focusing audits on the higher income taxpayers does not compensate for the smaller PDF bring about by income-level tax-payer-audit-selection method. The same pattern and conclusions can be drawn for PF = 1, except that in this case, the difference in CDI between strategic and random tax-payer-audit-selection is statistically significant (10,851,931 – 5,342,515 = 5,509,416, p-value of t-stat < .00001). As can be seen from Fig. X4, CDI for income-level attains the smallest CDI compared with strategic and random (10,851,931 – 4,131,763 = 6,720,168; 5,342,515 - 4,131,763 = 1,210,752; with both p-values of t-stats < .00001). As already explained for PF = 5, this happens because the bigger PDF of strategic and random compared with income-level offsets the smaller tax revenue that random and strategic might exhibit.

Table X4 shows an ANOVA for CDI, where we can see that its results are consistent with the previous analyses. The main effects of tax-payer-audit-selection and period-frauds are statistically significant as well as their interaction.

Table X4. ANOVA for Cumulative difference between real and reported income of taxpayers (CDI) corresponding to the full factorial experiment with 5 replications.

	Type III SS	df	MS	F	p-value
Corrected model	1.601×10^{14}	5	3.201×10^{13}	1435.61	< .0001
Intercept	1.022×10^{15}	1	1.022×10^{15}	45851.48	< .0001
tax-payer-audit-selection	8.360×10^{13}	2	4.180×10^{13}	1874.62	< .0001
period-frauds (PF)	2.638×10^{13}	1	2.638×10^{13}	1182.89	< .0001
tax-payer-audit-selection x PF	5.008×10^{13}	2	2.504×10^{13}	1122.97	< .0001
Error	5.351×10^{11}	24	2.230×10^{10}		
Total	1.183×10^{15}	30			

6. DISCUSSION

Aquí debemos usar la evidencia de resultados para discutirla sustantivamente y contestar las preguntas de investigación.

7. CONCLUSIONS

This paper uses agent-based modeling to investigate the conditional effectiveness of audit selection strategies for tax evasion detection under conditions of risk aversion and social influence through Moore neighborhood effects. Our primary finding is that **strategic selection based on Fiscal Risk Score (PRF) is conditionally effective**.

The key mechanism linking audit strategy to detection outcomes Is **risk aversion dynamics mediated by social learning memory**:

- **Short memory**: Agents cannot accumulate sufficient information about peer sanctions, so risk aversion remains heterogeneous and relatively low. Audit strategy selection becomes critical for maintaining compliance pressure.
- **Long memory**: Agents accumulate information about peer sanctions, so risk aversion converges toward high levels. Audit strategy selection becomes less critical because most agents are already highly compliant.

Policy Implications

5. **Audit strategy selection depends on social learning environment**: Tax administrations should consider the social learning environment when selecting audit strategies. In environments with limited social learning, strategic targeting is superior. In environments with strong social learning, strategic and random selection are approximately equivalent.
6. **Avoid predictable targeting patterns**: Predictable audit patterns create signaling effects that reduce compliance incentives for non-targeted taxpayers. These signaling effects can outweigh targeting efficiency gains.
7. **Trade-off between targeting efficiency and unpredictability**: Tax administrations face a fundamental trade-off between targeting efficiency (focusing audits on high-risk cases) and unpredictability (maintaining uncertainty about audit probability). The optimal strategy depends on the relative importance of these objectives and the social learning environment.
8. **Consider neighborhood effects**: Audit strategy effectiveness depends on the structure of social networks and the dynamics of social learning. Tax administrations should consider these factors when designing audit strategies.

Contributions to Literature

This paper contributes to the literature on tax evasion, audit strategy, and agent-based modeling by:

- **Demonstrating conditional effectiveness of risk-based targeting**: We show that risk-based targeting is conditionally effective, depending on social learning dynamics. This challenges conventional assumptions about audit strategy effectiveness.
- **Identifying risk aversion dynamics as a critical mechanism**: We identify how risk aversion adjustment through social learning mediates the relationship between audit strategy and detection outcomes.
- **Revealing predictability-deterrence trade-off**: We demonstrate that predictable targeting patterns can reduce overall deterrence effectiveness through signaling effects.
- **Integrating audit strategy with social learning**: We provide an integrated model that combines audit strategy selection with social learning dynamics and neighborhood effects.

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APPENDIX A: DETAILED STATISTICAL RESULTS

APPENDIX B: SENSITIVITY ANALYSIS